

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-GS-004

Title: Emergency Response Procedure for Gas Leak

Department: Safety and Environment

Revision: 04

Date: 15-Sep-2024

1. PURPOSE

To define the immediate response actions for all personnel upon detection of a gas leak at Bokaro Steel Plant, covering Blast Furnace Gas (BFG), Coke Oven Gas (COG), and Linz-Donawitz Gas (LDG).

2. SCOPE

Applicable to all areas where process gases are present. Mandatory for all plant personnel, contractors, and visitors.

3. GAS HAZARD CLASSIFICATION

- Blast Furnace Gas (BFG): CO content 20-28%. Highly toxic. Odourless.

- Coke Oven Gas (COG): Contains H₂, CH₄, CO. Flammable. Toxic.

- LD Gas (LDG): CO content 60-70%. Extremely toxic. Odourless.

CRITICAL: BFG and LDG are odourless. Never rely on smell for detection. Always carry a personal CO monitor.

4. DETECTION THRESHOLDS (REVISED)

- CO alarm (warning): 20 ppm

- CO alarm (evacuate): 35 ppm

- Explosive limit for COG: 6% in air (Lower Explosive Limit)

5. IMMEDIATE RESPONSE PROCEDURE

Step 1 — On Detection of Gas Leak

- Do NOT operate any electrical switches or use mobile phones in the affected area

- Raise the alarm using the nearest manual call point

- Call Emergency Control Room: Extension 100

Step 2 — Evacuation

- Evacuate all non-essential personnel immediately, moving upwind

- Minimum safe distance: 150 metres from the leak point (revised from 100 metres)

- Assemble at the department assembly point for head count

Step 3 — Isolate the Source (Trained Personnel Only)

- Only Gas Safety Officers wearing SCBA may attempt isolation

Step 4 — Notify

- Emergency Control Room: Extension 100

- Gas Safety Department: Extension 2310

6. RE-ENTRY PROTOCOL

- Re-entry only after written clearance from the Gas Safety Officer

- CO level must be below 20 ppm at all entry points

7. REVISION NOTE

Revision 04 lowers the evacuation threshold from 50 ppm to 35 ppm and increases the minimum safe distance from 100 to 150 metres.